

Assignment 1

1. Create 'spj' database'. (Database for suppliers, parts and jobs)
2. use spj database
3. Create following tables in spj database.
 - CREATE TABLE S (S# CHAR(5) NOT NULL, Sname CHAR(20), status SMALLINT, city CHAR(15));

```
kd1_99523_sandhya1> CREATE TABLE S(S# CHAR(5) NOT NULL,Sname CHAR(20),status SMALLINT,city CHAR(15));
Query OK, 0 rows affected (0.03 sec)

kd1_99523_sandhya1> INSERT INTO S VALUES('S1','Smith',20,'London');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO S VALUES('S2','Jones',10,'Paris');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO S VALUES('S3','Blake',30,'Paris');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO S VALUES('S4','Clark',20,'London');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO S VALUES('S5','Adams',30,'Athens');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> SHOW TABLES;
+-----+
| Tables_in_spj |
+-----+
| s              |
+-----+
1 row in set (0.00 sec)

kd1_99523_sandhya1> DESCRIBE S;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S#    | char(5) | NO   |     | NULL    |       |
| Sname | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
| city  | char(15) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

```
kd1_99523_sandhya1> DESCRIBE S;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S#    | char(5) | NO   |     | NULL    |       |
| Sname | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
| city  | char(15) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

kd1_99523_sandhya1> select * from s;
+-----+-----+-----+-----+
| S# | Sname | status | city |
+-----+-----+-----+-----+
| S1 | Smith | 20     | London |
| S2 | Jones | 10     | Paris |
| S3 | Blake | 30     | Paris |
| S4 | Clark | 20     | London |
| S5 | Adams | 30     | Athens |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

kd1_99523_sandhya1> |
```

CREATE TABLE P (P# CHAR(6) NOT NULL, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15));

```
kd1_99523_sandhya1> CREATE TABLE P(P# CHAR(6) NOT NULL,Pname CHAR(20),color CHAR(6),weight SMALLINT,city CHAR(15));
Query OK, 0 rows affected (0.03 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P1','Nut','Red',12,'London');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P2','Bolt','Green',17,'Paris');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P3','Screw','Blue',17,'Rome');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P4','Screw','Red',14,'London');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P5','Cam','Blue',12,'Paris');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO P VALUES('P6','Cog','Red',19,'London');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> select * from p;
+-----+-----+-----+-----+-----+
| P# | Pname | color | weight | city |
+-----+-----+-----+-----+-----+
| P1 | Nut   | Red   | 12      | London |
| P2 | Bolt  | Green | 17      | Paris  |
| P3 | Screw | Blue  | 17      | Rome   |
| P4 | Screw | Red   | 14      | London |
| P5 | Cam   | Blue  | 12      | Paris  |
| P6 | Cog   | Red   | 19      | London |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

- CREATE TABLE J (J# CHAR(4) NOT NULL, Jname CHAR(10), City CHAR(15));

```
kd1_99523_sandhya1> CREATE TABLE J(J# CHAR(4) NOT NULL,Jname CHAR(10),cityCHAR(15));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(C
15))' at line 1
kd1_99523_sandhya1> CREATE TABLE J (
->   'J#' CHAR(4) NOT NULL,
->   Jname CHAR(10),
->   city CHAR(15)
-> );
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J1','Sorter','Paris');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J2','Punch','Rome');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J3','Reader','Athens');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J4','Console','Athens');
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J5','Collator','London');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J6','Terminal','Oslo');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO J VALUES('J7','Tape','London');
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> select * from J;
+-----+-----+-----+
| J# | Jname | city |
+-----+-----+-----+
| J1 | Sorter | Paris |
| J2 | Punch  | Rome  |
| J3 | Reader | Athens |
| J4 | Console | Athens |
| J5 | Collator | London |
| J6 | Terminal | Oslo |
+-----+-----+-----+
```

- CREATE TABLE sp (S# CHAR(4) NOT NULL, P# CHAR(4) NOT NULL, J#CHAR(4) NOT NULL, QTY INT);

```
kd1_99523_sandhyal> CREATE TABLE sp('s#' CHAR(4) NOT NULL,'P#' CHAR(4) NOT NULL,'J#' CHAR(4) NOT NULL,QTY INT);
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S1','P1','J1',200);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S1','P1','J4',700);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J1',400);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J2',200);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J3',200);
Query OK, 1 row affected (0.04 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J4',500);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J5',600);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J6',400);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P3','J7',800);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S2','P5','J2',100);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S3','P3','J1',200);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S3','P4','J2',500);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S4','P6','J3',300);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S4','P6','J7',300);
Query OK, 1 row affected (0.00 sec)
```

```
kd1_99523_sandhyal> INSERT INTO sp VALUES('S4','P6','J7',300);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P2','J2',200);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P2','J4',100);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P5','J5',500);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P5','J7',100);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P6','J2',200);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P1','J4',100);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P3','J4',200);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P4','J4',800);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P5','J4',400);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhyal> INSERT INTO sp VALUES('S5','P6','J4',500);
Query OK, 1 row affected (0.01 sec)
```

```
kd1_99523_sandhyal> select * from sp;
```

S#	P#	J#	QTY
S1	P1	J1	200
S1	P1	J4	700
S2	P3	J1	400
S2	P3	J2	200
S2	P3	J3	200
S2	P3	J4	500
S2	P3	J5	600

```

kd1_99523_sandhya1> INSERT INTO sp VALUES('S5','P3','J4',200);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO sp VALUES('S5','P4','J4',800);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> INSERT INTO sp VALUES('S5','P5','J4',400);
Query OK, 1 row affected (0.00 sec)

kd1_99523_sandhya1> INSERT INTO sp VALUES('S5','P6','J4',500);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> select * from sp;
+----+----+----+----+
| s# | P# | J# | QTY |
+----+----+----+----+
| S1 | P1 | J1 | 200 |
| S1 | P1 | J4 | 700 |
| S2 | P3 | J1 | 400 |
| S2 | P3 | J2 | 200 |
| S2 | P3 | J3 | 200 |
| S2 | P3 | J4 | 500 |
| S2 | P3 | J5 | 600 |
| S2 | P3 | J6 | 400 |
| S2 | P3 | J7 | 800 |
| S2 | P5 | J2 | 100 |
| S3 | P3 | J1 | 200 |
| S3 | P4 | J2 | 500 |
| S4 | P6 | J3 | 300 |
| S4 | P6 | J7 | 300 |
| S5 | P2 | J2 | 200 |
| S5 | P2 | J4 | 100 |
| S5 | P5 | J5 | 500 |
| S5 | P5 | J7 | 100 |
| S5 | P6 | J2 | 200 |
| S5 | P1 | J4 | 100 |
| S5 | P3 | J4 | 200 |
| S5 | P4 | J4 | 800 |
| S5 | P5 | J4 | 400 |
| S5 | P6 | J4 | 500 |
+----+----+----+----+
24 rows in set (0.00 sec)

```

1. How many orders data is present in the database?

Ans : 10 records are stored in orders table.

2. How many customers are present in the database?

Ans: 7 Customers are present in database.

3. How many salespeople are present in the database?

Ans: 5 Peoples are present in database.

4. On which date order with highest amount is placed?

Ans: On 1990-10-04 highest amount order is placed.

5. For which order salesman got maximum commission? Hint: you need to refer data of two tables.

Ans: on order number of 3001, 3006 salesman got maximum commission of 0.15%.

6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.

Ans: No Salesman handling more than two customers every customer handling one or two customres only.

7. Which city have single customer?

Ans: Berlin city have only one customer.

8. Which city has multiple salespeople?

Ans: London city has mutiple salespeople.

9. What is name of customer and salesman of the maximum amount order?

Ans: Customer Cisneros and salesman Peel has maximum amount order.

10. Study relationship between the tables in sales database.

Ans : orders table is related to Customer table and salespeople by cnum (customer number)

and snum (salesman number)

orders - cnum and snum are foregin key.

Customer – snum is foregin key.

Order.cnum = customer.cnum

Order.snum = salespeople.snum